CTF Preparation Series

CTF Fundamentals (Network & Linux)

Links, slides and resources

gianemayumi.notion.site/ctf-series

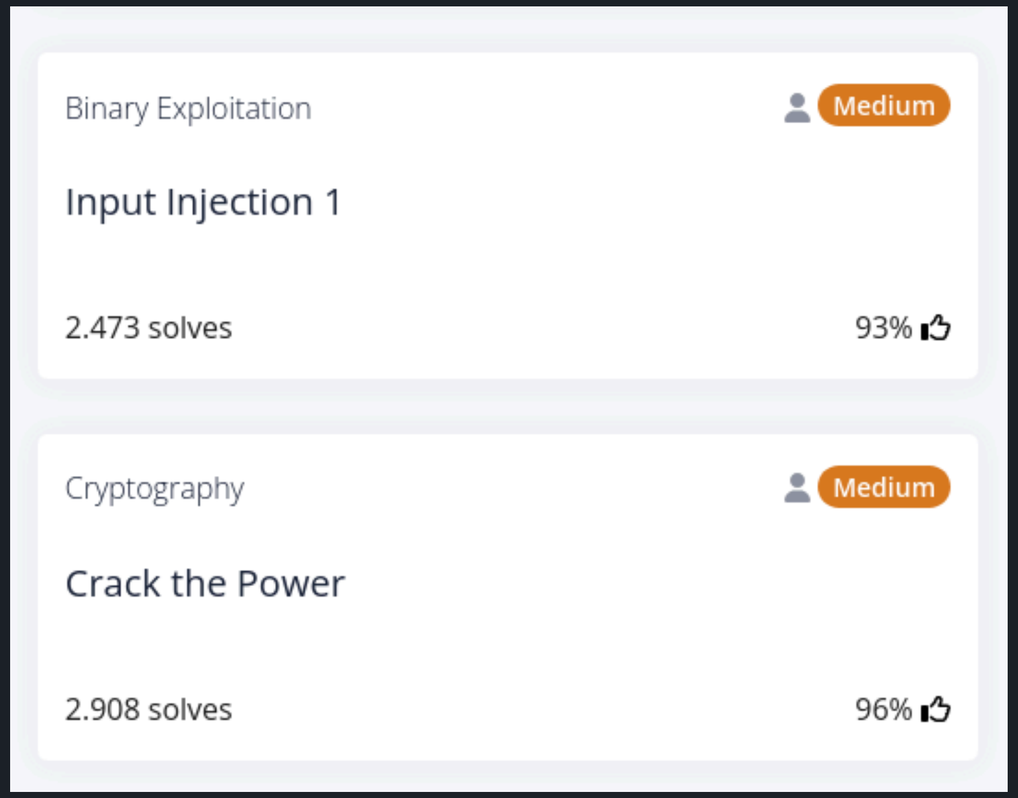
Create your account

https://play.picoctf.org/

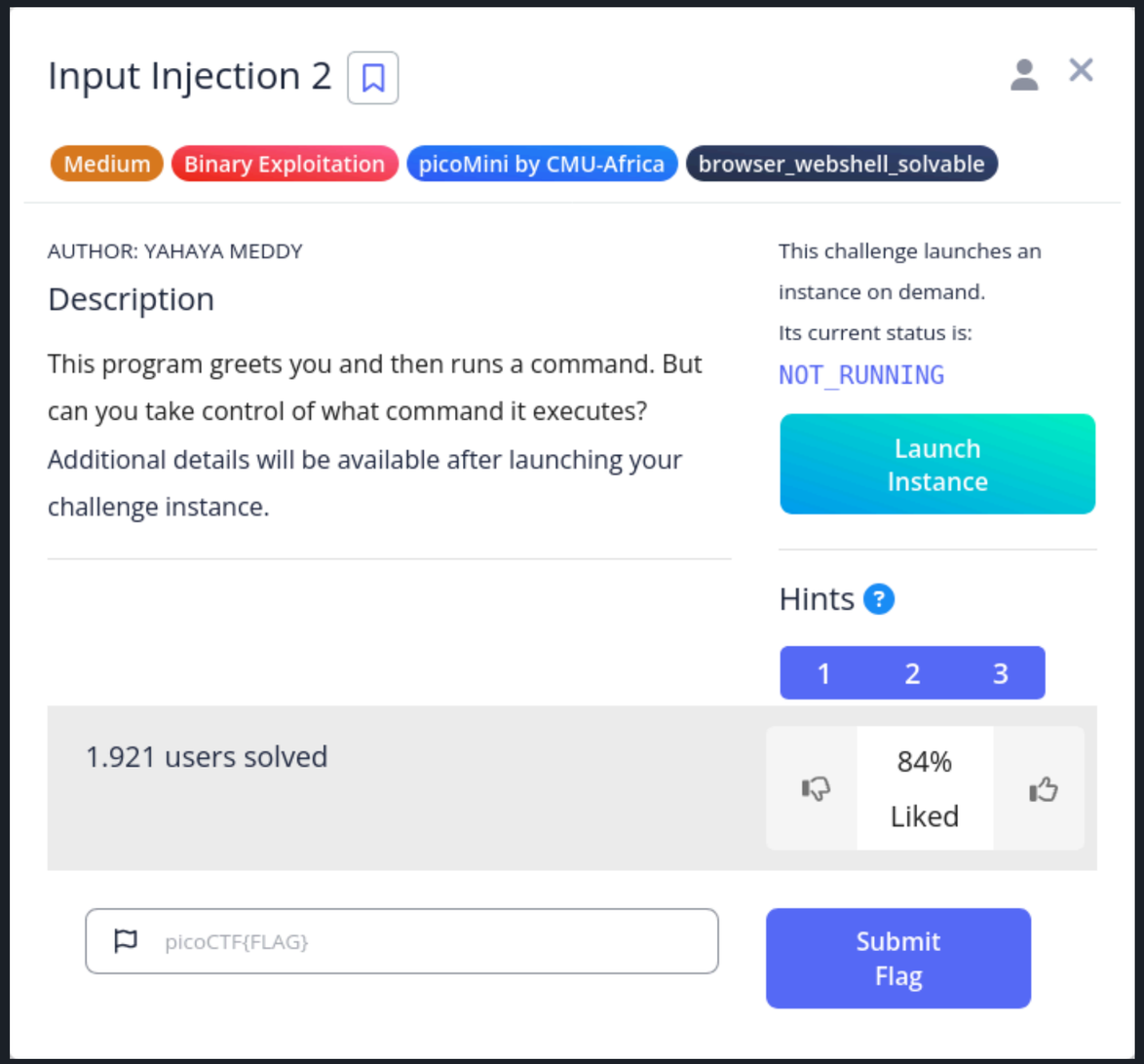
CTF

Capture The Flags (CTFs) are gamified competitive cybersecurity events that are based on different challenges or aspects of information security.

CTF structure



CTF structure



CTF structure

filter_	rank_	team_name_	quiz_pts_	flag_pts_	base_pts_	total_pts_
	1	XSLAY###	85	700	0	785
	2	TONY_###	75	550	0	625
	3	MAGDA###	65	580	0	625
	4	KONSI###	55	550	0	605
	5	PEPED###	55	500	0	555
	6	INVISI###	55	500	0	555
	7	NOK###	55	530	0	555
	8	SALVA###	45	330	0	365
	9	CASSI###	55	320	0	355

Learning resources

https://overthewire.org/wargames/

https://play.picoctf.org

https://tryhackme.com/

https://www.hackthebox.com/

Practice + write-ups

CTF Preparation Series

Expectations

Linux

Linux

WSL

Run a Linux environment on Windows

Dual Boot

Two different OS in the same computer

Kali Linux

Advanced Penetration Testing Linux distribution

VM

Software-based computer running Linux inside the computer

man

The system's manual pager.
Each page argument given to man is normally the name of a program, utility or function.

```
man [command]
```

Test it!

```
man ls
```

cat

Concatenate files and print on the standard output

```
cat [file]
```

Redirect

Method of controlling where the input and output of commands are directed.
It allows users to send command outputs to files or take inputs from files.

- >: Redirects standard output (overwrites file).
- >>: Redirects and appends output to a file.
- <: Takes input from a file instead of the keyboard.
- 2>: Redirects error messages to a file.
- &>: Redirects both standard output and error to a file.

https://www.geeksforgeeks.org/linux-unix/input-output-redirection-in-linux/

Pipe (|)

Send the output of one command/program/process to another command/program/process for further processing.

```
[command 1] | [command 2]
```

grep

Print lines that match patterns

```
[ command ] | grep “[pattern]”
```

Challenge 1

First Grep

Easy | General Skills | picoCTF 2019

AUTHOR: ALEX FULTON/DANNY TUNITIS

Description

Can you find the flag in the file? This would be really tedious to look through manually, something tells me there is a better way.

The flag is in this file.

Hints

1

109.486 users solved

92% Liked

picoCTF{FLAG}

Submit Flag

strings

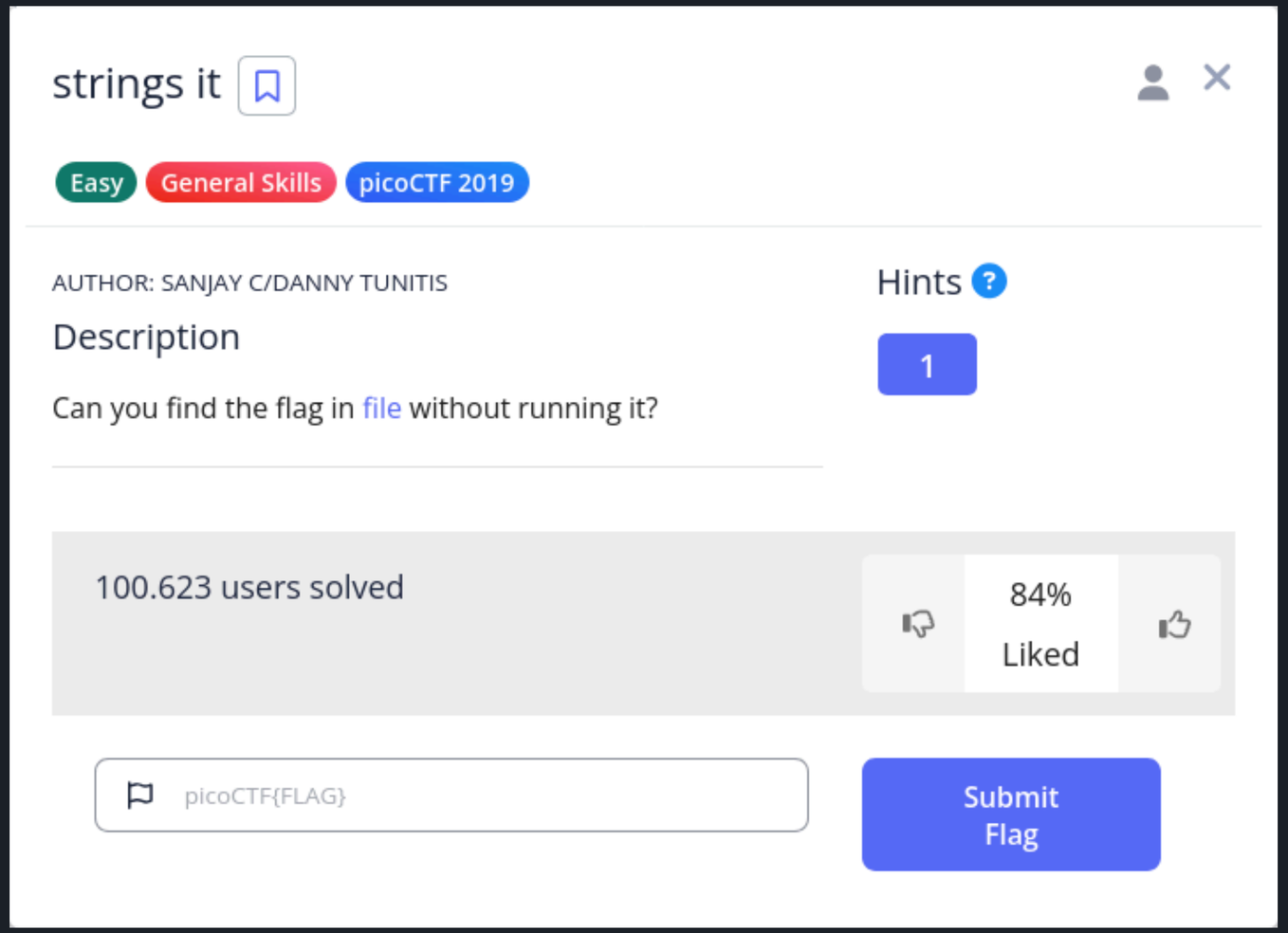
Print the strings of printable characters in files.

```
strings [ file ]
```

scripting

Bash and Python scripts to automate repetitive tasks

Challenge 2


strings it
Easy
General Skills
picoCTF 2019
AUTHOR: SANJAY C/DANNY TUNITIS
Description
Can you find the flag in file without running it?
Hints
1
100.623 users solved
84%
Liked
picoCTF{FLAG}
Submit Flag

Networking

ping

Sends out ICMP (Internet Control Message Protocol) echo request packets to the target host and waits for echo replies.

Test here!
google.com

https://phoenixnap.com/kb/linux-ping-command-examples

whois

Retrieve information about domain names, IP addresses, and network devices

Test here!
mcgill.ca

nmap

Tool that scans ports, status and services running.

Test here!
scanme.nmap.org

ncat

Ncat is a tool for reading, writing, redirecting, and encrypting information over a network. It supports a wide variety of security tests and can be used to interact with servers or as a backend for other tools.

Recommendation:
http://www.idc-online.com/technical_references/pdfs/data_communications/What_is_Netcat_and_How_to_use_it.pdf

ncat

Sinopse

```
ncat [OPTIONS...] [hostname] [port]
```

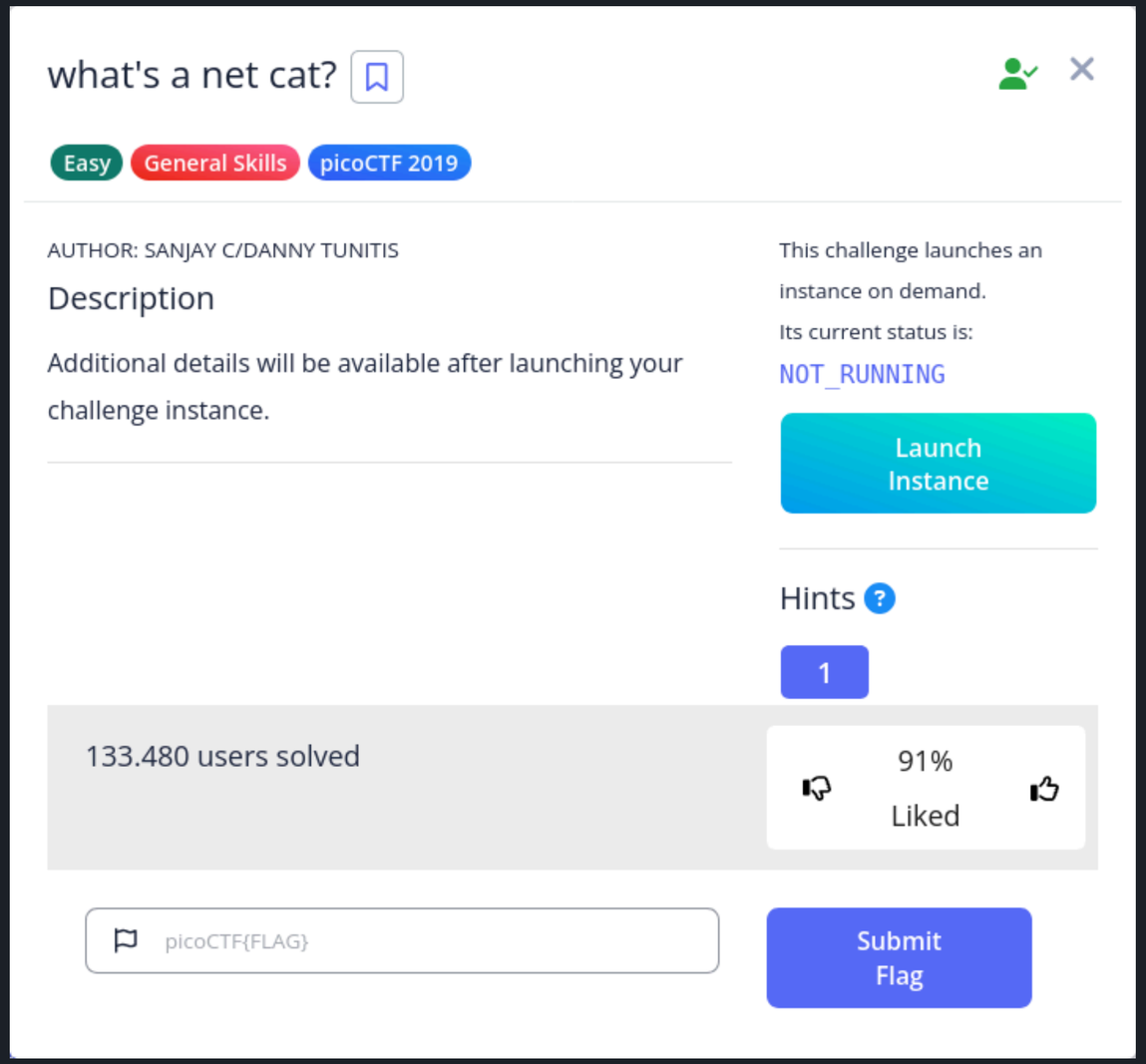
To listen in a specific port

```
ncat -l [port]
```

To connect to a remote port

```
ncat [hostname] [port]
```

Challenge 3


what's a net cat?
Easy
General Skills
picoCTF 2019
AUTHOR: SANJAY C/DANNY TUNITIS
Description
Additional details will be available after launching your challenge instance.
This challenge launches an instance on demand.
Its current status is:
NOT_RUNNING
Launch Instance
Hints
1
133.480 users solved
91%
Liked
picoCTF{FLAG}
Submit Flag

Challenge 4

plumbing

Medium | General Skills | picoCTF 2019

AUTHOR: ALEX FULTON/DANNY TUNITIS

Description

Sometimes you need to handle process data outside of a file. Can you find a way to keep the output from this program and search for the flag?
Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.
Its current status is:
NOT_RUNNING

Launch Instance

Hints

1 2

46.955 users solved

96% Liked

picoCTF{FLAG}

Submit Flag

Challenge 5

mus1c

Medium | General Skills | picoCTF 2019

AUTHOR: DANNY

Description

I wrote you a song. Put it in the picoCTF{} flag format.

Hints

1

32.229 users solved

31%

Liked

picoCTF{FLAG}

Submit Flag

Challenge 6

Glitch Cat

Easy | General Skills | Beginner picoMini 2022 | nc | shell | Python

AUTHOR: LT 'SYREAL' JONES

Description

Our flag printing service has started glitching!
Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.
Its current status is:
NOT_RUNNING

Launch Instance

Hints

1 2 3

59.275 users solved

78% Liked

picoCTF{FLAG}

Submit Flag

Challenge 7

Log Hunt

Easy | General Skills | picoMini by CMU-Africa | browser_webshell_solvable

AUTHOR: YAHAYA MEDDY

Description

Our server seems to be leaking pieces of a secret flag in its logs. The parts are scattered and sometimes repeated. Can you reconstruct the original flag?

Download the logs and figure out the full flag from the fragments.

Hints

1 2

22.032 users solved

95% Liked

picoCTF{FLAG}

Submit Flag

Thank you!

See you next week :)

Linux

https://overthewire.org/wargames/bandit/

TryHackMe Intro to Offensive Security

https://tryhackme.com/room/offensivesecurityintro

TryHackMe Intro to Networking

https://tryhackme.com/room/whatisnetworking